

Cell structure sample answers

SAMPLE PLACEHOLDER replace with your own answer key.

1. Nucleus: contains the DNA that controls the cell's activities.

Mitochondrion: site of aerobic respiration; releases ATP.

Ribosome: site of protein synthesis.

Plasma membrane: controls the movement of substances into and out of the cell.

2. Prokaryotic: no membrane-bound nucleus; circular DNA; 70S ribosomes.

Eukaryotic: has a membrane-bound nucleus; linear DNA with histones; 80S ribosomes.

4. Total magnification = $40 \times 10 = \times 400$.

$4 \text{ mm} / 400 = 0.01 \text{ mm} = 10 \text{ micrometres}$.